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SET-2

End Semester Examination 2025

Name of the Course: B. Tech. Biotechnology

Semester: 6th

Name of the Paper: Microbial Technology

Paper Code: TBT-604c

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Each sub question carries 10 marks

Q1	(20marks)	
(a)	What do you understand by the term xenobiotics. Explain the process of biodegradation of xenobiotics by microorganisms.	CO1
(b)	Discuss the role of microorganisms in human therapeutics and environment.	
(c)	What is microbial growth and microbial biomass. Define the steps involved in microbial degradation of hemicellulose.	
Q2	(20 marks)	
(a)	Describe the fermentation process used for xanthan gum production. What are the key nutrients required in the growth medium?	CO2
(b)	Explain the function of the A and B chains in insulin, and how are they produced and combined in recombinant systems?	
(c)	Discuss the advantage and disadvantage of Genetically Modified Crops.	
Q3	(20 marks)	
(a)	Implement the knowledge of recombinant DNA technology in production of hGH.	CO3
(b)	A seed company wants to develop a new Bt crop targeting a different pest. What steps would you recommend in the research and development process?	
(c)	Explain the term antibiotics? Describe the industrial production of any one antibiotic with its applications.	
Q4	(20 marks)	
(a)	Explain the role of <i>Xanthomonas campestris</i> in xanthan gum production. Why is this bacterium specifically used?	CO4
(b)	Draw the well labelled diagram of Hepatitis B virus and give diagrammatic representation of its vaccine production.	
(c)	Summarize different categories of vaccines with suitable examples.	
Q5	(20 marks)	
(a)	Compare submerged fermentation (SmF) and solid-state fermentation (SSF) for citric acid production. Which method would you recommend for large-scale industrial production in a developing country, and why?	CO5
(b)	Vaccine hesitancy remains high for newer technologies like mRNA. Propose and justify strategies to improve public trust in rapidly developed vaccines without compromising safety.	
(c)	Compare different natural and artificial gene transfer methods.	

